

AI/ML Internship – Day 48 Notes & Tasks

Module 6: Custom Object Detection – Training YOLO on Your Own Dataset

Part 1: Recap of Day 47

Yesterday we learned:

-  Video Object Detection
-  Real-Time Webcam Detection
-  OpenCV Integration
-  Live Object Detection

Today we will learn:

-  Custom Object Detection
 -  Training YOLO with Our Own Dataset
-

Part 2: What is Custom Object Detection?

Until now, YOLO could detect:

- Person
- Car
- Dog
- Cat
- Bicycle

because it was already trained on these objects.

Problem

Suppose we want to detect:

- Company ID Card
- Company Logo
- Safety Helmet

- Machine Parts
- Fruits
- Kerala Vehicle Number Plates

YOLO does not know these objects by default.

Solution

Train YOLO using our own dataset.

This is called:

Custom Object Detection

Part 3: Real-World Examples

Manufacturing

Detect:

- Defective Products
 - Damaged Components
-

Safety Monitoring

Detect:

- Helmet
 - Safety Vest
-

Vehicle Systems

Detect:

- Number Plates
 - Vehicle Types
-

Healthcare

Detect:

- Tumors
 - Medical Equipment
-

📌 Part 4: Dataset

What is a Dataset?

A collection of images used for training.

Example:

Helmet Dataset

helmet1.jpg

helmet2.jpg

helmet3.jpg

helmet4.jpg

📌 Part 5: Dataset Components

For every image:

image.jpg

we need:

image.txt

Example:

helmet1.jpg

helmet1.txt

📌 Part 6: What is Annotation?

Annotation means drawing bounding boxes around objects.

Example:

```
-----  
|  HELMET  |  
-----
```

and saving its coordinates.

📌 Part 7: Annotation Tools

Popular annotation tools:

Labelling

Easy

Beginner Friendly

Roboflow

Online Tool

Most Popular

CVAT

Advanced Tool

📌 Part 8: Dataset Structure

YOLO expects:

dataset/

└─ images

| └─ train
| └─ val

└─ labels

 └─ train
 └─ val

📌 Part 9: Example Dataset

dataset/

images/

```
└─ train/
|  helmet1.jpg
|  helmet2.jpg
```

```
└─ val/
   helmet3.jpg
```

labels/

```
└─ train/
|  helmet1.txt
|  helmet2.txt
```

```
└─ val/
   helmet3.txt
```

Part 10: Data.yaml File

YOLO requires:

data.yaml

Example:

train: dataset/images/train

val: dataset/images/val

nc: 1

names:

- helmet

Part 11: Understanding data.yaml

train

Training images location.

val

Validation images location.

nc

Number of classes.

Example:

Helmet

Only one class.

nc: 1

names

Class names.

names:

- helmet

Part 12: Install YOLO

pip install ultralytics

Part 13: Import YOLO

from ultralytics import YOLO

Part 14: Load Pretrained Model

model = YOLO("yolov8n.pt")

Part 15: Train Custom Dataset

from ultralytics import YOLO

model = YOLO("yolov8n.pt")

```
model.train(  
    data="data.yaml",  
    epochs=10  
)
```

Part 16: Understanding Training Parameters

data

Dataset configuration.

```
data="data.yaml"
```

epochs

Number of training cycles.

```
epochs=10
```

Meaning:

Train entire dataset 10 times.

Part 17: Training Workflow

Dataset



Annotation



data.yaml



YOLO Training



New Model



Custom Detection

Part 18: Output After Training

YOLO creates:

runs/

└─ detect/

train/

Inside:

best.pt

Part 19: What is best.pt?

This is your trained model.

Example:

helmet_detector.pt

YOLO saves it as:

best.pt

Part 20: Use Trained Model

```
from ultralytics import YOLO
```

```
model = YOLO(  
    "runs/detect/train/weights/best.pt"  
)
```

```
results = model("helmet_test.jpg")
```

```
results[0].show()
```

Part 21: Example Output

Input:

Worker Wearing Helmet

Output:

Helmet → 96%

with bounding box.

Part 22: Theory Questions

What is Custom Object Detection?

Training YOLO for new objects.

What is Annotation?

Drawing bounding boxes and labeling objects.

What is data.yaml?

Dataset configuration file.

What is Epoch?

One complete training cycle.

What is best.pt?

Final trained YOLO model.

Part 23: Common Mistakes

Wrong Folder Structure

 Images and labels not matching.

Missing Annotation Files

 Image exists.

 TXT file missing.

Wrong data.yaml

 Incorrect paths.

Very Small Dataset

 Poor performance.

Part 24: Real-World Industry Projects

Helmet Detection

Construction sites.

Face Mask Detection

Hospitals.

Vehicle Number Plate Detection

Traffic systems.

Product Detection

Retail stores.

Defect Detection

Factories.

Tasks for Day 48

Theory

1. What is Custom Object Detection?
 2. What is Annotation?
 3. What is data.yaml?
 4. What is an Epoch?
 5. What is best.pt?
-

Practical Task 1

Install:

```
pip install ultralytics
```

Practical Task 2

Research:

- Labellmg
 - Roboflow
 - CVAT
-

Practical Task 3

Create a sample dataset structure.

Practical Task 4

Write a sample:

data.yaml

for Helmet Detection.

Practical Task 5

Run:

```
model.train(  
    data="data.yaml",  
    epochs=10  
)
```

Understand training process.